LAB 1
Digital Electronics 2B
EEDE316
Semester 1
Luther xabisile Ndondo

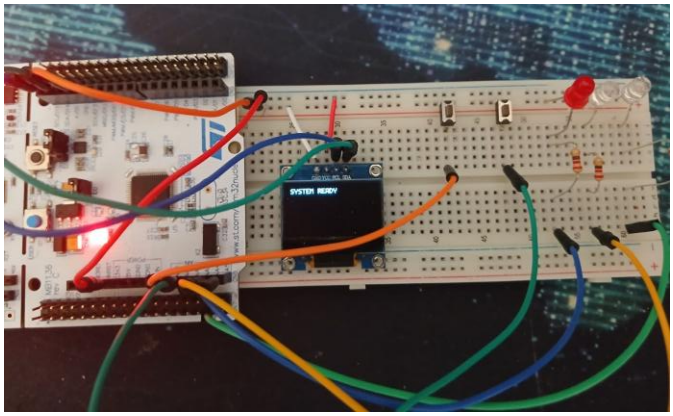
Introduction:

The objective of this project is to interface pushbuttons and LEDs with the STM32F401RE Nucleo board. The system must also display status messages on an OLED display using the Serial Peripheral Interface (SPI) communication protocol.

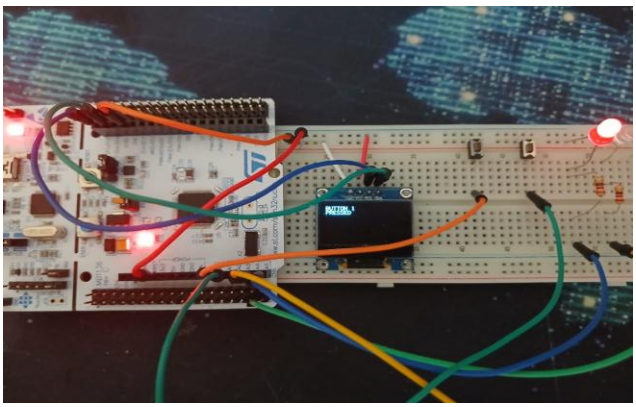
Components Required:

- STM32F401RE Nucleo board
- 2 × Pushbuttons (normally open)
- 3 × LEDs
- 5 × 220 Ω resistors
- 0.96” OLED Display (SSD1306, SPI interface)
- Breadboard • Jumper wires

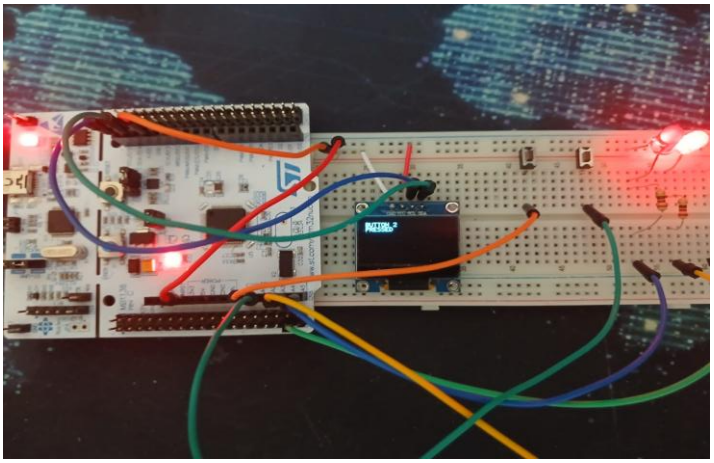
System Ready



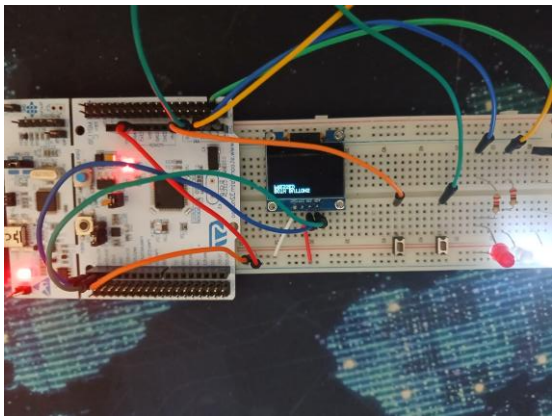
Button 1 Pressed



Button 2 Pressed



Both Button Pressed



```c
while (1)
{
    uint8_t btn1 = (HAL_GPIO_ReadPin(GPIOA, BTN2A1_Pin) == GPIO_PIN_RESET);
    uint8_t btn2 = (HAL_GPIO_ReadPin(GPIOA, BTN2_Pin)   == GPIO_PIN_RESET);

    if (btn1 && btn2)
    {
        HAL_GPIO_WritePin(GPIOC, LED1_Pin|LED2_Pin, GPIO_PIN_RESET);
        HAL_GPIO_WritePin(GPIOC, LED3_Pin, GPIO_PIN_SET);
        OLED_Clear();
        OLED_ShowString(0, "BOTH BUTTONS");
        OLED_ShowString(1, "PRESSED");
        HAL_Delay(5000);
        HAL_GPIO_WritePin(GPIOC, LED3_Pin, GPIO_PIN_RESET);
        OLED_Clear();
        OLED_ShowString(0, "SYSTEM READY");
    }
    else if (btn1)
    {
        HAL_GPIO_WritePin(GPIOC, LED2_Pin|LED3_Pin, GPIO_PIN_RESET);
        HAL_GPIO_WritePin(GPIOC, LED1_Pin, GPIO_PIN_SET);
        OLED_Clear();
        OLED_ShowString(0, "BUTTON 1");
        OLED_ShowString(1, "PRESSED");
        HAL_Delay(5000);
        HAL_GPIO_WritePin(GPIOC, LED1_Pin, GPIO_PIN_RESET);
        OLED_Clear();
        OLED_ShowString(0, "SYSTEM READY");
    }
    else if (btn2)
    {
        HAL_GPIO_WritePin(GPIOC, LED1_Pin|LED3_Pin, GPIO_PIN_RESET);
        HAL_GPIO_WritePin(GPIOC, LED2_Pin, GPIO_PIN_SET);
        OLED_Clear();
        OLED_ShowString(0, "BUTTON 2");
        OLED_ShowString(1, "PRESSED");
        HAL_Delay(5000);
        HAL_GPIO_WritePin(GPIOC, LED2_Pin, GPIO_PIN_RESET);
        OLED_Clear();
        OLED_ShowString(0, "SYSTEM READY");
    }

    HAL_Delay(50);
}
```